

Lecture Notes on

Quantum Mechanics

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These notes cover quantum mechanics at the undergraduate level. The sources used are:

- Griffiths: *Introduction to Quantum Mechanics*
- Sakurai: *Modern Quantum Mechanics*

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1 Fundamentals

1.1 The Stern-Gerlach Experiment

We'll begin by demonstrating the inadequacies of classical theory in its failure to explain the results of the Stern-Gerlach experiment.

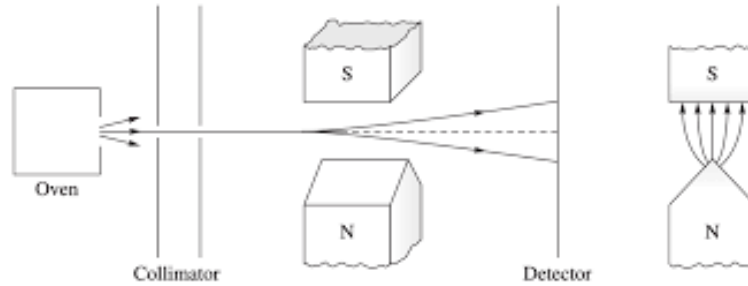


Figure 1: The Stern-Gerlach apparatus

- Silver atoms contain 47 electrons, 46 of which form a spherically symmetric electron cloud with no angular momentum. The 47th electron produces no orbital angular momentum so all angular momentum of the atom is due to the electron's spin. Thus we have the magnetic moment is proportional to the electron spin $\mu \propto \mathbf{S}$.
- The atom is then heated in an oven and sent through a magnetic field $\mathbf{B}$ in the z direction. The interaction energy is then $-\mu \cdot \mathbf{B}$ so the force it experiences is $F = \mu_z \frac{\partial B_z}{\partial z}$.
- Since the atom is randomly oriented when leaving the oven, we'd expect a continuous distribution of possible trajectories depending on the amount of spin lying on the z axis. However, we observe the trajectories splitting into two distinct components, such that μ can only be identified with two possible values of the z component of $\mathbf{S}$, $S_z = \hbar/2$ or $S_z = -\hbar/2$. This allows us to deduce that electron spin angular momentum is quantized.
- Suppose we block off the negative S_z beam, and then send this through another Stern-Gerlach apparatus with magnetic field in the x direction. We observe that 50% of the particles separate into the positive x beam and the other 50% separate the negative x beam. Finally, selecting either resulting beam through a final apparatus in the z direction, it splits off into two equal beams! This appears impossible classically because the first apparatus blocked all atoms with negative z spin yet we still observe them after measuring their x spin.
- This means the atoms entering the third beam cannot simultaneously have S_z and S_x spin simultaneously, which allows us to conjecture that we cannot determine both S_x and S_z simultaneously. In fact, measuring the spin in the x direction destroys previous information about the z direction.

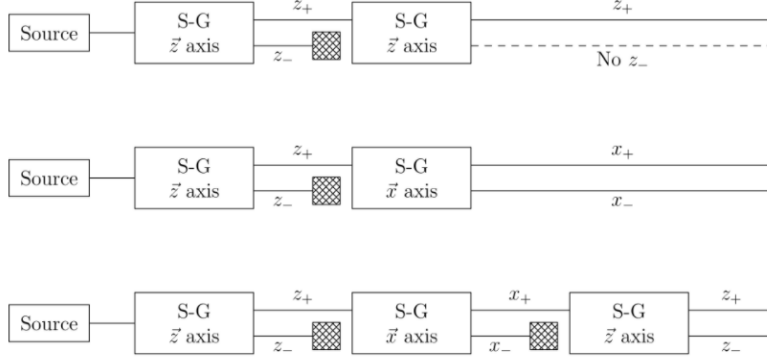


Figure 2: Several Stern-Gerlach apparatus chained together

We can make an analogy of this behavior with the polarization of light. If we sent light through an x filter and then a y filter, no light passes through. However if we place in between an x' filter oriented at an angle of 45° from the x direction in the xy plane, light passes through the y filter! Even though the x filter destroyed all y polarized light initially, the x' filter destroyed all previous information of that.

- We first conjecture that there is a correspondance between angular momentum and light polarization as follows:

$$S_z \pm \text{atoms} \leftrightarrow \text{x, y polarized light}$$

$$S_x \pm \text{atoms} \leftrightarrow \text{x', y' polarized light}$$

- The electric field of x polarized light is $\mathbf{E} = E_0 \hat{\mathbf{x}} \cos(kz - \omega t)$ and likewise for y polarized. Then x' and y' polarized light can be expressed as $\mathbf{E}_{\mathbf{x}'} = \frac{1}{\sqrt{2}} \mathbf{E}_{\mathbf{x}} + \frac{1}{\sqrt{2}} \mathbf{E}_{\mathbf{y}}$ and $\mathbf{E}_{\mathbf{y}'} = -\frac{1}{\sqrt{2}} \mathbf{E}_{\mathbf{x}} + \frac{1}{\sqrt{2}} \mathbf{E}_{\mathbf{y}}$
- We represent the “state” of positive x spin using a ket in Dirac notation $|S_x; +\rangle$. Using the correspondance above, we might represent the positive and negative spin states as:

$$|S_x; +\rangle = \frac{1}{\sqrt{2}} |S_z; +\rangle + \frac{1}{\sqrt{2}} |S_z; -\rangle$$

$$|S_x; -\rangle = -\frac{1}{\sqrt{2}} |S_z; +\rangle + \frac{1}{\sqrt{2}} |S_z; -\rangle$$

The atoms coming out of the positive x spin filter are in a superposition of the S_z+ and S_z- states so when sending them through the final filter, we observe both cases even though the negative spin should’ve been “filtered out” already.

- By symmetry, decomposition of S_y into S_z components should be very similar, but we’ve already used all possible linear combinations of S_z kets. The idea instead is to make a correspondance with circularly polarized light, where the y component is out of phase with the x component: $\mathbf{E} = E_0 \left[\frac{1}{\sqrt{2}} \hat{\mathbf{x}} \cos(kz - \omega t) + \frac{1}{\sqrt{2}} \hat{\mathbf{y}} \cos(kz - \omega t + \frac{\pi}{2}) \right]$. We can use complex notation by writing $\text{Re}(\epsilon) = \mathbf{E}$. So for right circularly polarized light we have

$$\epsilon = E_0 \left[\frac{1}{\sqrt{2}} \hat{\mathbf{x}} e^{i(kz - \omega t)} + \frac{i}{\sqrt{2}} \hat{\mathbf{y}} e^{i(kz - \omega t)} \right]$$

- Now by correspondance, we can express the spin in the y direction as

$$|S_y; \pm\rangle = \frac{1}{\sqrt{2}} |S_z; +\rangle \pm \frac{i}{\sqrt{2}} |S_z; -\rangle$$

The main idea is that restricting ourselves to the real numbers is not expressive enough to be able to decompose both S_y and S_x into S_z components. So our ket vectors necessarily belong to some complex vector space.

1.2 Bra-ket Notation

Now we will lay down some formalism for Dirac notation. We postulate that a state ket $|\alpha\rangle$ contains all the information about a physical state. Kets are elements of a Hilbert space meaning all the properties of vector spaces apply.

- Multiplying a ket by a scalar $c|\alpha\rangle$ represents the same physical state. Only the direction is significant.
- Elements of the dual space are denoted as bras $\langle\alpha|$, with a one to one correspondence with elements in the ket space:

$$\begin{aligned} |\alpha\rangle &\leftrightarrow \langle\alpha| \\ |\alpha\rangle + |\beta\rangle &\leftrightarrow \langle\alpha| + \langle\beta| \end{aligned}$$

Meanwhile, under scalar multiplication, the dual vector uses the complex conjugate

$$c|\alpha\rangle \leftrightarrow c^* \langle\alpha|$$

- We denote the inner product of a bra and a ket as $\langle\alpha|\beta\rangle$. Note the inner product is not symmetric in its arguments:

$$\langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*$$

A consequence of this is that $\langle\alpha|\alpha\rangle$ is necessarily real.

- A vector's inner product with itself is strictly positive

$$\langle\alpha|\alpha\rangle \geq 0$$

The equality holds iff $|\alpha\rangle$ is a null ket. This is equivalent to saying the metric is positive definite.

Real world values like position and momentum are also known as observables. An observable can be represented by an operator A on the vector $A|\alpha\rangle$. An eigenket obeys $A|a\rangle = a|a\rangle$ where a is a real scalar value. In this case, $|a\rangle$ is what's called an eigenket. The physical state corresponding to an eigenket is an eigenstate. For example, the spin 1/2 eigenstates obey $S_z|S_z; \pm\rangle = \pm\frac{\hbar}{2}|S_z; \pm\rangle$. Some further properties of observables:

- Two observables are equal iff $X|\alpha\rangle = Y|\alpha\rangle$ for any ket. We call X a null operator of $X|\alpha\rangle = 0$ for any ket.

- Observables are both commutative and associative under addition

$$X + Y = Y + X$$

$$X + (Y + Z) = (X + Y) + Z$$

All observables except for the time reversal operator are linear:

$$X(c|\alpha\rangle + d|\beta\rangle) = cX|\alpha\rangle + dX|\beta\rangle$$

- The dual correspondance of operators acting on kets/bras is:

$$X|\alpha\rangle \leftrightarrow \langle\alpha|X^\dagger$$

where $X^\dagger$ is the Hermitian adjoint of X . An operator is Hermitian if

$$X = X^\dagger$$

- Operators are generally not commutative under multiplication, but are associative. The Hermitian adjoint of a product is

$$(XY)^\dagger = Y^\dagger X^\dagger$$

which can be shown using the dual correspondance.

- The **associative axiom** states that any legal multiplication with kets, bras and operators is associative. For example, consider the outer product $|\beta\rangle\langle\alpha|$. If we were to act this on a ket, $(|\beta\rangle\langle\alpha|)|\gamma\rangle$, by the associative axiom we can rewrite this as $|\beta\rangle(\langle\alpha|\gamma\rangle)$ which is just a scalar times a ket. Thus, the outer product is an operator.
- By the associative axiom, we have the equality $\langle\beta|(X|\alpha\rangle) = (\langle\beta|X)|\alpha\rangle$. Since there is no distinction between the two, we just drop the parenthesis and write $\langle\beta|X|\alpha\rangle$. Using the property of the inner product,

$$\langle\beta|X|\alpha\rangle = \langle\alpha|X^\dagger|\beta\rangle^*$$

Hermitian operators are of special importance to us. An especially important theorem is that the eigenvalues of Hermitian operators are real, and eigenkets corresponding to different eigenvalues are orthogonal.

Suppose we have the dual eigenvectors $A|\alpha\rangle = a|\alpha\rangle$ and $\langle\beta|A = \langle\beta|b^*$, Multiplying the first equation by $\langle\beta|$ on the left and the second by $|\alpha\rangle$ on the right and subtracting, we have $(a - b^*)\langle\beta|\alpha\rangle = 0$. Suppose $a = b$. Then we have $a = a^*$ so the eigenvalues must be real. Suppose $a \neq b$. Then because we've shown the reality condition, the difference $a - b^*$ cannot vanish, so to satisfy the equation, the inner product $\langle\beta|\alpha\rangle = 0$ must vanish. So different eigenkets must be orthogonal.

Some important consequences of this result:

- We conventionally normalize the eigenkets to form an orthonormal set. Additionally, the set of eigenkets is complete ie; it spans the entire ket space.

- Any arbitrary ket $|\alpha\rangle$ can be expanded in eigenkets $|\alpha\rangle = \sum_i c_i |a_i\rangle$ where the coefficients are $c_i = \langle a_i|\alpha\rangle$. This means the expansion is $\alpha = \sum_i |a_i\rangle \langle a_i|\alpha\rangle$. But by the associative axiom, we can factor the $|\alpha\rangle$ out, giving us

$$\sum_i |a_i\rangle \langle a_i| = 1$$

which is known as the **completeness relation**. 1 is known as the identity operator. This fact is very useful because we can place it in the middle of any equation.

Example. Inserting the completeness relation between $\langle\alpha|\alpha\rangle$ we have $\langle\alpha|(\sum_i |a_i\rangle \langle a_i|)|\alpha\rangle = \sum_i |\langle a_i|\alpha\rangle|^2$. But if $|\alpha\rangle$ is normalized, then the coefficient expansion satisfies $\sum_i |c_i|^2 = \sum_i |\langle a_i|\alpha\rangle|^2 = 1$

- The outer product $|a_i\rangle \langle a_i|$ is also known as the **projection operator**, also denoted as $\Lambda_{a_i} \equiv |a_i\rangle \langle a_i|$. The completeness relation can also be written as $\sum_i \Lambda_{a_i} = 1$. This makes sense as the sum of all projections should return the original vector, so in this way this formulation of a projection is more “natural”.
- By using the completeness relation twice, we can express an operator $X = \sum_i \sum_j |a_i\rangle \langle a_i| X |a_j\rangle \langle a_j|$ as a matrix. The components of the matrix are $\langle a_i|X|a_j\rangle$ where i indexes the row and j indexes the columns. Note that the kets correspond to column vectors and bras correspond to row vectors by duality. We also already have that $\langle a_i|X|a_j\rangle = \langle a_j|X^\dagger|a_i\rangle^*$. But if X is hermitian, then this gives us

$$\langle a_i|X|a_j\rangle = \langle a_j|X|a_i\rangle^*$$

meaning X is its own conjugate transpose, which is the more colloquial definition of Hermitian adjoint.

2 The Schrödinger Equation

2.1 Wave Functions

Classically, we work with masses and attempt to solve for their equations of motions $x(t)$ using corresponding Newton’s laws or energy equations. The quantum mechanics approach is different; the idea is to instead solve for a particle’s wave function $\Psi(x, t)$ and from there we can do anything. The goal then is to solve the Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

After solving the Schrödinger equation for some initial wave function $\Psi(x, 0)$ the equation actually gives us all future values of the wave function $\Psi(x, t)$.

The statistical interpretation of the wave function goes as follows:

- Born’s statistical interpretation of the wave function states that

$$\int_b^a |\Psi(x, t)|^2 dx$$

is the probability of finding the particle at $a \leq x \leq b$ at time t . Note that while Ψ is a complex valued function, $|\Psi|^2 = \Psi\Psi^*$ where Ψ^* is the complex conjugate of Ψ is always real and nonnegative.

- An obvious consequence is this is that even if you know everything about a particle (ie; you know its wavefunction), you still cannot predict with certainty the outcome of an experiment to measure its exact position.
- Before measurement, a particle's position is not well defined. Only after measurement can we say the particle has a "position," and by doing so we collapse the particle's wavefunction to a dirac delta centered around the measured position. This is known as the **Copenhagen Interpretation**
- The integral over all possible positions must be equal to one, since the particle must be located somewhere.

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

If we have a wavefunction, we need to make sure its normalized for it to be physical. We can normalize it to obey the integral by multiplying it by a scale factor. A glance at the Schrödinger equation one can see that if $\Psi(x, t)$ is a solution, then $A\Psi(x, t)$ is also a solution for nonzero constant A .

- One can also show that the normalization integral does not change with time, so there is no need to constantly renormalized solutions as the particle evolves with time.
- If we interpret $|\Psi(x, t)|^2$ to be the probability, then we can say the expectation value of position is

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx$$

Note that the interpretation of this quantity is the expected value of the position from measuring particles with wave function Ψ . Simply remeasuring the same particle over again will not work, since the first measurement will collapse the particle into some other wave function with exact position

- By differentiating the expected position with respect to time, one can obtain the expectation value of the velocity. It's customary for us to work using momentum so we have

$$\langle p \rangle = -i\hbar \int \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) dx$$

- A more illustrative way to write the expectation values for position and momentum are:

$$\langle x \rangle = \int \Psi^* \cdot x \cdot \Psi dx$$

$$\langle p \rangle = -i\hbar \int \Psi^* \cdot \left(\frac{\partial}{\partial x} \right) \cdot \Psi dx$$

We say the the x operator "represents" position and the $-i\hbar \frac{\partial}{\partial x}$ operator "represents" momentum.

- All classical quantities can be represented as a function of position and angular momentum. Then the representation of a classical quantity $Q(x, p)$ can be written as

$$\langle Q(x, p) \rangle = \int \Psi^* Q \left(x, -i\hbar \frac{\partial}{\partial x} \right) \Psi dx.$$

- **Ehrenfest's Theorem** states that expectation values obey their classical laws. For example, the time derivative of momentum is

$$\frac{d\langle p \rangle}{dt} = \langle -\frac{\partial V}{\partial x} \rangle$$

2.2 Time Independent Schrödinger Equation

Let us assume that the potential $V(x)$ is independent of time. Then using separation of variables, we can substitute $\Psi(x, t) = \psi(x)\phi(t)$ into the Schrödinger equation giving us $i\hbar\frac{1}{\phi}\frac{d\phi}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V$. Then for the equality to hold, both sides must be equal to some constant value E leaving us with two ordinary differential equations:

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar}\phi$$

$$-\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V = E$$

The second equation is the **Time Independent Schrödinger Equation**. The first equation defines the characteristic time dependence and can be solved trivially $\phi(t) = e^{-iEt/\hbar}$. Separable solutions obey several important properties that we're interested in:

- These solutions are **stationary states**. What this means is while the wave function is obviously time dependent, $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, the probability density is not:

$$|\Psi(x, t)|^2 = \Psi\Psi^* = \psi^*e^{+iEt/\hbar}\psi e^{-iEt/\hbar} = |\psi(x)|^2$$

We can further show that the expectation value of any classical quantity is also not time dependent in a stationary state:

$$\langle Q(x, p) \rangle = \int \psi^* Q\left(x, -i\hbar\frac{\partial}{\partial x}\right) \psi dx.$$

- The constant E represents the Hamiltonian of the system. Classically, the Hamiltonian is $H(x, p) = \frac{p^2}{2m} + V(x)$ so the corresponding Hamiltonian operator is $\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$. So the time independent Schrödinger equation can be rewritten as

$$\hat{H}\psi = E\psi$$

Then the expectation values of the Hamiltonian is

$$\langle H \rangle = \int \psi^* \hat{H} \psi dx = \int \psi^* E \psi dx = E$$

$$\langle H^2 \rangle = \int \psi^* \hat{H}^2 \psi dx = \int \psi^* E (\hat{H} \psi) dx = \int \psi^* E^2 \psi dx = E^2$$

This means the standard deviation is 0. So any ensemble of stationary states with the same wavefunction have their Hamiltonian measured to be the exact same E .

- Every solution to the Schrödinger equation can be written in the form of a sum of stationary solutions.

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-E_n t / \hbar}$$

So we can basically decompose any wave function into the sum of wavefunctions associated with specific energy levels. The quantity $|c_n|^2$ represents the probability that a measurement of the energy will return value E_n , so we also must demand

$$\sum_{n=1}^{\infty} |c_n|^2 = 1$$

$$\langle H \rangle = \sum_{n=1}^{\infty} |c_n|^2 E_n$$

2.3 Bound States

Consider the potential function given by

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & \text{otherwise} \end{cases}$$

Outside the potential well, $\psi(x)$ must be 0 because $V(x)$ goes to infinity. So we have $\psi(x) = 0$ for $x < 0$ and $x > a$. Inside the well, the Schrödinger equation gives us $-\frac{\hbar}{2m} \frac{d^2 \psi}{dx^2} = E \psi$. The solution is just the simple harmonic oscillator with $k = \sqrt{2mE}/\hbar$. So the general solution is $\psi(x) = A \sin(kx) + B \cos(kx)$. Using the boundary condition that the wavefunction must go to 0 at $x = 0$ and $x = a$, we obtain

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right)$$

with corresponding energy

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

Some properties of the ψ_n functions:

- ψ_n alternates parity around the center of the well. ψ_1 , the ground state, is even, ψ_2 , the first excited state is odd, ψ_3 is even and so on.
- Each ψ_n are orthonormal. Namely:

$$\int \psi_m(x) * \psi_n(x) dx = \delta_{mn}$$

- They form a complete basis for any function $f(x)$:

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi}{a} x\right)$$

This is simply a fourier series for $f(x)$, and this can represent any function $f(x)$ by **Dirichlet's Theorem**. To actually determine the coefficients, multiply both sides by ψ_m^* and then integrate

$$\int \psi_m(x)^* f(x) dx = \sum_{n=1}^{\infty} c_n \int \psi_m(x)^* \psi_n(x) dx = \sum_{n=1}^{\infty} c_n \delta_{nm} = c_m$$

$$c_n = \int \psi_n(x)^* f(x) dx$$

Now consider the potential inside a harmonic oscillator $V(x) = \frac{1}{2}kx^2$. This potential is actually very illustrative because any general potential $V(x)$ can be taylor expanded into $V(x) \approx V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \dots$ but notice that we can simplify by working with x_0 as an equilibrium point x_0 , so the $V'(x_0)$ term drops, and we can also choose to define V up to a scalar constant such that $V(x_0) = 0$. If we assume $x - x_0$ stays small, then we could describe any oscillatory motion with small amplitude.

The Schrödinger equation is now

$$-\frac{\hbar}{2m} \frac{d^2\psi}{dx^2} = \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

where $\omega = \sqrt{k/m}$. To solve this we introduce the **ladder operators** $\hat{a}_-$ and $\hat{a}_+$

- We define the operator $\hat{a}_{\pm}$ as

$$\hat{a}_{\pm} \equiv \frac{1}{\sqrt{2\hbar m\omega}} (\mp i\hat{p} + m\omega x)$$

The product is $\hat{a}_- \hat{a}_+ = \frac{1}{2\hbar m\omega} [\hat{p}^2 + (m\omega x)^2] - \frac{i}{\hbar} (\hat{x}\hat{p} - \hat{p}\hat{x})$. Here, $(\hat{x}\hat{p} - \hat{p}\hat{x}) \equiv [\hat{x}, \hat{p}]$ where $[\hat{x}, \hat{p}] = i\hbar$ is what's known as the **commutator** of $\hat{x}$ and $\hat{p}$.

- Using the ladder operators, notice that $[\hat{a}_-, \hat{a}_+] = 1$ and we can rewrite the Hamiltonian and Schrödinger equation as

$$\hat{H} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

$$\hbar\omega \left(\hat{a}_{\pm} \hat{a}_{\mp} \pm \frac{1}{2} \right) \psi = E\psi$$

- The $\hat{a}_+$ and $\hat{a}_-$ are accordingly named the **raising** and **lowering** operators. If ψ is a solution with energy E , then $\hat{a}_+\psi$ is also a solution with energy $E + \hbar\omega$ and $\hat{a}_-\psi$ is a solution with energy $E - \hbar\omega$. Notice that at the ground state, we are unable to lower the wave function any further, so we set $\hat{a}_-\psi_0 = 0$. Solving this differential equation gives us the ground state and energy:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} x^2}$$

$$E_0 = \frac{1}{2}\hbar\omega$$

- Then we can find the n th excited states by repeatedly applying the raising/lowering operators:

$$\hat{a}_+\psi_n = \sqrt{n+1}\psi_{n+1}$$

$$\hat{a}_-\psi_n = \sqrt{n}\psi_{n-1}$$

- It may also be helpful to represent the position and momentum operators using the ladder operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}_+ + \hat{a}_-)$$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}_+ - \hat{a}_-)$$

You could also solve the Schrödinger's equation analytically using a power series, and obtain

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}$$

where $\xi = \sqrt{\frac{m\omega}{\hbar}}x$ and H_n are the **Hermite Polynomials**.

2.4 Scattering States

In the examples above, we have the total energy is less than the potential energy at the boundaries, so solutions are bound states. However, we can also have states where the particle enters from infinity into a finite potential, and then leaves to infinity again. Solutions to these cases are called scattering states.

A particle in free space isn't subject to any forces so $V(x) = 0$. This differential equation is much easier to solve, with stationary solutions

$$\Psi_k(x, t) = A e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)}$$

where $k \equiv \pm \frac{\sqrt{2mE}}{\hbar}$. A positive k means the wave travelling to the right, while a negative one is traveling to the left. The wave travels with phase velocity

$$v_{\text{quantum}} = \frac{\hbar|k|}{2m} = \sqrt{\frac{E}{2m}} = \frac{1}{2}v_{\text{classical}}$$

It's important to distinguish the group velocity from the phase velocity. The group velocity is the speed at which the "envelope" of superpositions travels at which represents the motion of the particle. Here, $v_{\text{group}} = \frac{d\omega}{dk} = v_{\text{classical}}$ so this problem is resolved. However, another problem is that this wave function is not even normalizable!

- To resolve this issue, instead of summing over separable solutions, we integrate over the variable k :

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(k) e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)} dk$$

- We can use **Plancherel's theorem** to determine $\psi(k)$ from the initial conditions:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx$$

2.5 The Hydrogen Atom

In 3 dimensions, Hamiltonian operator is now $\frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V$. Notice that the momentum operator now becomes the gradient $\mathbf{p} = -i\hbar\nabla$. Thus:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V \Psi$$

However, most of the time the potential we will work with is spherically symmetric, in that it depends only on distance from the radius $V(r)$. Here it makes most sense to use spherical coordinates. By separation of variables, let

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

The angular equation gives us

$$\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] = -\ell(\ell + 1)$$

Performing separation of variables again with $Y(\theta, \phi) = \Theta(\theta)\Phi(\phi)$ we now have

$$\begin{aligned} \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) + \ell(\ell + 1) \sin^2 \theta &= m^2 \\ \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} &= -m^2 \end{aligned}$$

The ϕ equation is easy to solve. The solution is just $\Phi(\phi) = e^{im\phi}$. Note that we also demand $\Phi(\phi + 2\pi) = \Phi(\phi)$ which implies m must be integer. Meanwhile, the solution to the θ equation is

$$\Theta(\theta) = AP_\ell^m(\cos(\theta))$$

where P_ℓ^m are the **Legendre Functions** defined as:

$$P_\ell^m(x) \equiv (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x)$$

and P_ℓ are the **Legendre Polynomials**. Thus, the solution to the angular equation are the **Spherical Harmonics**:

$$Y_\ell^m(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m)!}{(\ell + m)!}} e^{im\phi} P_\ell^m(\cos \theta).$$

To solve the radial equation, instead we have $V(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$. Solving this differential equation is genuinely so terrible so we'll skip that. The solution is

$$\psi_{nlm} = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}} e^{-r/na} \left(\frac{2r}{na}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2r}{na}\right) Y_l^m(\theta, \phi)$$

where $L_{q-p}^p(x) \equiv (-1)^p \left(\frac{d}{dx}\right)^p L_q(x)$ is the associated **Laguerre polynomial**. Here, $L_q(x) \equiv e^x \left(\frac{d}{dx}\right)^q (e^{-x} x^q)$ is the q th Laguerre polynomial. It's quite nasty, but generally it's extremely rare for the Schrödinger equation to even have a closed form solution like this. Generally speaking, we won't have this luxury with systems like the helium atom.

The allowed energies are

$$E_n = - \left[\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n^2} \right]$$

with the lowest possible energy state and

$$a \equiv \frac{4\pi\epsilon_0\hbar^2}{me^2}$$

is the **Bohr Radius**, the smallest possible orbit.

3 Symmetries and Conservation Laws